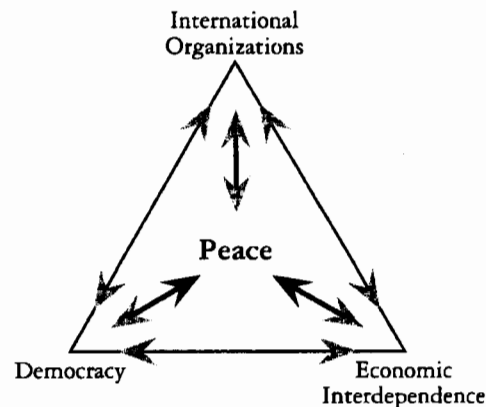


*from Triangulating Peace
by Russett*

Source: RUSSETT, BRUCE, AND JOHN R. ONEAL. (2001) *Triangulating Peace: Democracy, Interdependence, and International Organizations*. New York: Norton.

Both Democracy and Economic Interdependence Reduce Conflict



The results we reported in Chapter 3 indicate that democracy is a strong force for international peace. Yet we should resist the temptation to accept that evidence as conclusive until we consider at least one other influence: commercial relations that create a high degree of economic interdependence. Perhaps what looks like the effect of democratic institutions and culture is really the result of the interdependence that arises most naturally between states with open economies. Almost all democracies, though varying to a degree in the role played by the state, have capitalist economic systems that involve extensive competition in free markets among economic agents, including those in other countries. Consequently, democracies tend to trade extensively with one another. Because of the correlation between democracy and open markets, we need to reconsider the consequences of democracy through tests in which we control for the influence of interdependence.

There are three possibilities. The first is that the apparent benefits of democracy for interstate relations are mostly, or even entirely, the effect

of economic interdependence. The second is that economic interdependence and peace are both consequences of democracy, so that consideration of interstate commerce adds nothing to our understanding of the causes of war; it is only democracy that is important. The third possibility, the one most consistent with the Kantian perspective, is that democracy and economic interdependence make independent contributions: both constrain the use of force by creating powerful incentives for peace. In this case, those pairs of states that are both democratic and economically interdependent would tend to be more peaceful than pairs that are either only democratic or only interdependent. This chapter seeks to determine which of these three possibilities is correct. Here we expand our examination of the democratic peace to consider the effects of economic interdependence on the likelihood of conflict.

In the liberal view, trade and foreign investment, as well as the institutions and practices of democratic governance, should reduce the incidence of militarized disputes between countries. This thesis, like the liberal account of the peacefulness of democracies, has its origins in a classical literature that most clearly addressed the individual characteristics of states, arguing that democratic states and those economically open will be more peaceful in their diplomatic and military relations with others. But consideration of the pacific benefits of trade, as with the study of the democratic peace, has progressed through careful attention to the conflict-reducing potential of interdependence in dyadic analyses. Indeed, the effects of either democracy or interdependence on states' behavior can be obscured unless careful attention is paid to their bilateral consequences.¹ Consequently, we begin the empirical analyses reported below by examining the effects of economically important bilateral trade. We consider as well, however, whether a country's openness to trade with all countries, not just its relations with the other member of a dyad, also reduces the likelihood of conflict. Together, these analyses mirror our earlier interest in the peaceful relations that occur when both members of a dyad are democratic and the finding that democracies are generally more peaceful than autocracies.

¹The importance of analyzing the behavior of pairs of states is all the more evident when the liberal peace is evaluated within the wider context of international relations theory. Realism stresses the importance of alliances, relative power, and geostrategic location—influences best represented dyadically.

Our analyses here develop and build on those of the previous chapter. To reflect our attention to the second of the Kantian variables, the graphic at the head of this chapter includes a second dark arrow, from the lower right corner of the Kantian triangle to the center. The principal question of this chapter is whether trade makes an independent contribution to the probability that a pair of states will experience a militarized dispute. We also want to know if democracy, alliances, and the balance of power continue to be important influences when our measures of the economic importance of trade are added to our explanatory model. The tests we report below allow us to compare the relative benefits of the liberals' political and economic prescriptions for peace. Other questions, notably whether economic growth also contributes to peace—or provides the means for the exercise of power—are also explored.

If trade does appear to contribute to peace, we need to consider the reciprocal relation. Perhaps economic interdependence is largely a consequence, rather than a cause, of the absence of international disputes. After all, commercial agents can be expected to avoid the risks and costs of international violence where possible, and trading with the enemy is usually prohibited or extensively regulated by national governments. There may well be an association between trade patterns and conflict, but it may be primarily conflict that affects commerce, rather than the other way round. This chapter will not fully resolve that question, but it provides a good start, and we shall return to this issue later in the book.

The Liberal Peace: Classical Perspectives and Recent Research

KEY SECTION:

The classical liberals advocated policies to increase liberty and prosperity. They sought to empower the commercial class politically and to abolish royal charters, monopolies, and the protectionist policies of mercantilism so as to encourage entrepreneurship and increase productive efficiency. They also expected democracy and free-market economics to diminish the frequency of war. Political scientists have addressed the connection between democracy and international conflict in recent years, but they have shown less interest in the consequences of free trade and economic interdependence. Yet, expanded trade was advocated as a remedy for war even before democracy was a realistic possibility in most countries.

In the early seventeenth century, Emeric Crucé concluded that wars

arose from international misunderstandings and the domination of society by the warrior class and that both could be reduced by expanding commerce: trade would create common interests and increase the prosperity and political power of the peaceful, productive members of society. Similarly for Kant,

The *spirit of commerce* sooner or later takes hold of every people, and it cannot exist side by side with war. And of all the powers (or means) at the disposal of the state, *financial power* can probably be relied on most. Thus states find themselves compelled to promote the noble cause of peace, though not exactly from motives of morality. And wherever in the world there is a threat of war breaking out, they will try to prevent it by mediation. (Kant [1795] 1970, 114; italics in original)

Kant's view is especially interesting because of when and where he wrote. He lived in Königsberg, a city in Prussia in the late eighteenth century. (Königsberg is now a part of Russia and is called Kaliningrad.) Königsberg was no democracy, but was part of a hereditary monarchy. International law was not extensively developed at the time, and there were essentially no international organizations. But Kant knew firsthand about the political effects of foreign commerce, for Königsberg had long been an independent trading state and was involved in economic exchange with much of Europe. In thinking about the causes of peace, he embedded trade within the right to "hospitality," or temporary sojourn, whereby foreigners could carry on the peaceful pursuit of their livelihoods while temporarily on the territory of other states. Königsberg had been a member of the Hanseatic League during its heyday from the thirteenth to the seventeenth centuries. The Hanseatic League was a loose confederation of independent, self-governing trading towns, devoted to promoting and protecting its trading monopolies in northern Europe. It was able to coordinate economic sanctions or even war against nonmembers, and it had sufficient dispute-settling mechanisms to avoid wars among its members (Lister 1999, chap. 3).

François Quesnay, Anne Robert Turgot, and the French Physiocrats; Adam Smith, David Ricardo, and John Stuart Mill, in England; and Thomas Paine, in the United States, were other theoreticians or political figures who emphasized the role of economic relations in promoting peace (Howard 1978). In the mid-nineteenth century, Richard Cobden and the Manchester school of "commercial liberalism" argued that the

cost of war made it anachronistic, as did Norman Angell and Joseph Schumpeter somewhat later. In Cobden's terms, "Besides dictating the disuse of warlike establishments, free trade (for of that beneficent doctrine we are speaking) arms its votaries by its own pacific nature, in that eternal truth—the more any nation traffics abroad upon free and honest principles, the less it will be in danger of wars" (1886, 222; italics in original). Over time, trade also encouraged the development of international law and organizations, because these were needed to regulate and manage commercial relations, just as similar institutions were necessary in domestic economies.

The free traders—especially of nineteenth-century Britain—had an economic interest in the ideology they promoted. They sought a commercial republic of the world, of economic attraction rather than political rule (Semmel 1970), because economic interdependence was thought to create transnational ties that encouraged accommodation rather than conflict. In this way, economic interdependence reinforces the pacific benefits of democratic institutions and norms. Thus, material incentives add their force to law and morality. The benefits of interdependence are also central to functionalist accounts of political integration in Europe (Mitrany 1966) and are reminiscent of some socialists' emphasis on the virtues of internationalism (Domke 1988, 43–51).

Despite this impressive intellectual pedigree, the role of economic interdependence in preventing conflict has until recently been neglected (Levy 1989a). The benefits of trade may not be symmetrical and may favor the side with the stronger economic power in the market, but trade is always to some degree a mutually beneficial interaction; otherwise, it would not be undertaken. This gives each party a stake in the economic well-being of the other—and in avoiding militarized disputes. The nineteenth-century liberal argument derived primarily from a view that individuals act rationally in accordance with their economic interests. It is hardly in a state's interest to fight another if its citizens sell their goods, obtain imports (raw materials, capital goods, intermediate products, or consumer goods), or have financial investments or investors there. If my factory is located in your country, bombing your industry means, in effect, bombing my own property. Of course, trade can be redirected, at least over time, by political leaders who see the clouds of war on the horizon. But goods and services from alternative suppliers would cost more and/or be inferior in quality, and shifting exports means competing with

existing suppliers elsewhere, lower prices, and less profit. Indeed, the need to switch to the second-best trading partner may involve such high costs that a state is seriously vulnerable to a disruption of trade (Keohane and Nye 1977, 8–13).

Many interest groups, economic and otherwise, compete for the attention of government leaders and seek to influence their choices. Some of these interest groups will not care much about foreign policy, and some may even have economic interests linked to international conflict (for instance, arms manufacturers or those whose products could, in times of conflict, be sold in place of goods currently imported). The political influence of domestic economic interests will always be most important; this is especially true in large countries, which tend to be less dependent upon international trade. Yet if commerce between two countries constitutes a substantial share of the national income of one or both, important commercial interests will have a political stake in maintaining good relations. Commerce increases the power of special interest groups that benefit from foreign trade and investment. This includes not just bankers and the captains of industry but also their workers, consumers, suppliers, and the whole network of secondary economic beneficiaries: the automobile dealer who sells cars to those who work in factories that export abroad, those workers' grocers and restaurant owners, and many others. Rational political leaders want to stay in office, and to do so they must respond to the demands of constituents who are economically and, hence, politically powerful. If maintaining trade is important to continuing national prosperity and growth, leaders will be responsive to its beneficiaries. They may also see this as an important element of national security.

A somewhat different argument linking economic interdependence and peace focuses on the role of trade and foreign investment as media for communicating on a broad range of matters beyond the specific commercial exchanges that take place. Trade exposes a state's citizens to the ideas and perspectives of citizens of other countries on a wide range of issues (Lerner 1956; Russett 1963; Rosecrance 1986). These communications, too, form an important channel for averting interstate conflict. Economic interdependence contributes to the construction of a "security community" (Deutsch et al. 1957), in which shared values make the resort to force unimaginable. Common values create a shared sense of identity, which Deutsch calls "we-feeling." Very likely, both rationalist and constructivist arguments are correct. One influence may be stronger than the

other in one situation or another, but ultimately they are complementary.

Democracy, with its commitment to individual liberty, may reduce conflict not just directly but indirectly by encouraging interdependence. In democracies, economically powerful groups are likely to be politically powerful as well (Papayanou 1996). Political and economic freedoms allow individuals to form transnational associations that may be able to influence policy (Verdier 1994; Risse-Kappen 1995a). Trade agreements among democracies may be particularly long lasting. Because executives in democratic countries must persuade and accommodate other powerful groups—the legislature, their political party, interest groups, the public—they may be more likely to abide by their international commitments than nondemocratic leaders, whose power is less subject to checks and balances. Consequently, democracies should be better at promoting and sustaining interdependence because economic ties require credible commitments regarding the terms of trade, regulation of capital flows, and the adjudication of contractual disputes (Martin 2000). As we shall see later, democracies are indeed inclined to trade with one another.

A challenge to the liberal view comes from those who emphasize that economic ties not only offer the prospect of mutual gain but may also transmit economic ills and create rivalry over the division of benefits. In the seventeenth century, England and Holland—both major trading states—fought heavily over colonial territory and access to foreign markets. Some analysts of the age of Western imperialism, such as the liberal J. A. Hobson (1902) and the Marxist revolutionary V. I. Lenin ([1916] 1929), vigorously developed such arguments. Lenin, in fact, regarded imperialism as the "highest stage of capitalism." He considered imperialism an inevitable consequence of the growth of monopolies in capitalist economies and believed that it would lead to such terrible wars among the great powers, all of whom were scrambling for economic gain abroad, as to destroy the capitalist system and open the door for socialism worldwide.

More recently, critics of capitalism have shifted their attention away from the potential for conflict between imperial powers and focused instead on the likelihood of conflict between a powerful state and a much smaller economic partner. When the political, military, and economic power of states is vastly different, dependency theorists suggest, trade and investment create not interdependence but dependency. This disproportionately benefits the larger country because it is able to use its power to

manipulate the terms of exchange. In this view, too, trade can be a source of conflict (Hirschman 1945; Keohane and Nye 1977; Kroll 1993): asymmetrical economic relations lead to dependency, exploitation, and militarized disputes (dos Santos 1970; Robinson 1976; Mearsheimer 1992). If a small economy is dominated by a big one, popular resentment may bubble over and lead to violence. Or a revolutionary government may seize great agricultural estates, mineral assets, or factories owned by nationals of the great power, provoking military threats or retaliation. Consequently, conflict can arise because the weaker party resists what it sees as unfair treatment and exploitation or because the stronger state seeks to enforce its advantage. Some analysts doubt, therefore, that trade makes an important contribution to world peace, at least between states of greatly unequal size.

It is certainly possible to find historical examples that support dependency theorists' view. Something like this happened in Cuba after Fidel Castro's revolution in 1960. Castro was committed to a program of socialism, which entailed large-scale expropriation of private property and of Americans' property in particular, because they owned much of the agricultural land that was used to produce sugar, Cuba's principal export. Castro not only nationalized foreign holdings but he also switched Cuba's pattern of trade. He bartered Cuban sugar—no longer given a privileged position in the U.S. market—to the Soviet Union in exchange for oil, which had previously been imported from the United States. The result was a rapid deterioration in relations between the United States and Cuba, leading to a series of economic sanctions. In 1961, the United States sponsored the disastrous Bay of Pigs invasion by Cuban exiles. Castro then openly allied himself with the Soviet Union, accepting the Soviet weapons that prompted the Cuban missile crisis of 1962. The possibility of disputes arising out of asymmetrical economic relations must be considered in any systematic examination of the consequences of trade for interstate conflict.

However, despite some examples to the contrary, recent social scientific research has generally concluded that trade does reduce conflict.² Nor is the evidence for this beneficial effect limited to the post-World War II era. Way (1997) found strong support, in both the nineteenth and twen-

²Polachek 1980; Gasiorowski and Polachek 1982; Gasiorowski 1986; Polachek 1992; Polachek and McDonald 1992.

tieth centuries, for the pacific benefits of bilateral trade as well as democracy. Similarly, Domke (1988) reported that countries with high levels of total exports relative to the size of their economies were less likely to initiate wars than countries that were relatively self-sufficient. Indeed, he found that countries that were open to the global economy were more peaceful than democratic states that were not open. Looking at the international system as a whole over nearly two centuries, Mansfield (1994) found that a high level of world trade reduced the number of wars initiated in the subsequent five years. One recent review, while cautioning that "outstanding empirical and theoretical questions" make "elevating this hypothesis to the status of a social scientific 'law' . . . premature," concludes that "the position advocated by liberalism is strongly supported by the existing literature."³

Analytical Problems

In accord with the findings of most—though not all—social scientific studies regarding this link, we, too, have found that economically important trade reduces interstate conflict. In previous articles, we reported the results of analyses using different sets of cases, over different periods, and with a variety of measures of interdependence. Below we report new empirical analyses that provide additional support for this conclusion. First, however, we must address some important analytical issues.

One involves how the effect of trade is to be assessed. If trade is to affect national decision makers, its beneficiaries must be politically important. They will be more influential, as we suggested above, when trade is more important to the national economy. For instance, the value of France's exports to Belgium is, allowing for shipping costs, reporting errors, and some technicalities, identical to the value of Belgium's imports from France. Likewise, France's imports from Belgium are the same as Belgium's exports to France. But since France's total economy is six times larger than Belgium's, this trade is more important economically to Belgium than it is to France. Consequently, Belgium's leaders are more subject to the political influence exercised by the economic actors who

³McMillan 1997, 34; for more recent reviews, see Oneal and Russett 1999a; Barbieri and Schneider 1999.

benefit from this exchange. Therefore, unlike Barbieri (1996), in assessing the impact of trade on the decisions of national leaders, we must calculate the value of trade relative to the size of a nation's economy, usually as measured by its gross domestic product (GDP). Presumably, the smaller country will have more incentive to avoid the costs of conflict with the bigger country than vice versa.

In assessing the effects of trade on dyadic relations, it is also important to control for the distance separating two countries. As common sense and the evidence of the previous chapter tell us, countries' ability to fight one another depends heavily on how near they are. Distance limits their ability to exert effective military power against one another. Trade, too, is clearly affected by distance. Distance increases the costs of trade, even today when technological innovation has dramatically reduced the expense of shipping by air or sea; distance influenced economic relations much more fifty or a hundred years ago. As a result, countries that are far apart trade less with each other than do close ones, all other things being equal (Tinbergen 1962; Deardorff 1995). Because proximate states tend both to fight more and to have higher levels of trade, if we are not careful, we might conclude that trade increases the likelihood of conflict. This would be a mistake, and it is a common one when analysts (e.g., Gaddis 1987, 224–25) note that states that have high levels of mutual trade (and, not coincidentally, are contiguous) often fight each other. To estimate the benefits of trade properly, we must control for the effect of distance on the frequency of militarized disputes. The right question is: With distance held constant, is trade positively or negatively related to the probability that two states will fight? In other words, of two states that are equally distant, which is more likely to become a country's military adversary, the one with which there is a high level of trade or the one with which commercial relations are minimal?

It is especially important to control for distance when investigating the effect of trade. There is little need to worry that distance is an important influence on whether two states are both democratic and on whether they are likely to fight. Two countries can be democratic even if they are on opposite sides of the globe, though to a degree, democracy does seem to be "contagious" among neighbors. But in investigating the effect of trade on conflict, not only do we run the risk of finding a correlation between high levels of trade and conflict if we fail to control for distance, we could also observe many "false negatives," where we mistakenly attribute the

lack of conflict between, say, Iceland and China to their negligible economic ties. Much of the confusion about the benefits of trade arises from not recognizing the importance of distance.

In testing the liberal thesis regarding the pacific benefits of interdependence, it is also important to think carefully about the direction of causation. Economically important trade is expected to reduce military conflict because conflict would disrupt this trade and impose significant economic costs on those involved. Thus, the theory argues not only that trade will limit the use of force but also that conflict will reduce the level of trade. Indeed, it is because militarized disputes are likely to reduce trade and adversely affect economic interests that interdependence is expected to lower the chance that disputes will occur.

There are two reasons why this feedback of reciprocal relation between the two variables should occur. The first is simple enough: traders are assumed to be rational economic actors. They will stop trading if their goods (or their lives) are endangered by military hostilities. At a minimum, they will seek a greater profit margin to compensate for the greater risk, but these higher costs will lower demand and reduce commerce. Indeed, traders may see war clouds on the horizon and, out of self-interest, limit their commercial activity so as to cut their risks. Thus, trade may fall in response to conflict, and it may fall in anticipation of it. Accordingly, declining trade may signal deteriorating political relations. It may even exacerbate interstate tensions by reducing the flow of valuable communications or heightening the alarm of national leaders, who may be encouraged to turn to military action to limit or reverse the economic consequences of the lost trade (Copeland 1996).

Conflict will also influence the level of trade because states usually take steps to prevent their citizens from "trading with the enemy." Commerce between states at war was not uncommon in earlier centuries (Barbieri and Levy 1999), but modern states are more effective in limiting such activity. States can impose a variety of economic sanctions to weaken or punish their adversaries, including a complete embargo on trade and investment. States can also act against countries with which they are not actually at war; for example, by limiting the sale of goods that are considered strategic: key raw materials or technological products with possible military applications. Many restrictions of this type were imposed on East-West trade during the cold war, and there are still significant limitations on U.S. trade with China, Russia, and other countries. The United

States also maintains a virtual embargo on trade with Cuba, Iran, Iraq, Libya, North Korea, and Serbia. Indeed, many types of military equipment can only be sold to members of the NATO alliance, Israel, and a few other countries that have a "special relationship" with the United States.

These are all examples of how conflict or governments' fear of conflict can lead to policies that reduce the level of trade with another state, but states can also act to increase their economic interdependence with those with which they have good relations. Various political scientists, such as Pollins (1989a, 1989b), Gowa and Mansfield (1993), and Gowa (1994), have shown that trade is influenced by states' security interests. A state is apt to trade more with an ally because it need not fear that the economic gains that arise from their commercial relations will be used to threaten its security. After World War II, the United States deliberately opened its markets to Japan to facilitate that country's revival as a prosperous and democratic ally. Consequently, we need to look at whether interdependence has important pacific benefits when the influence of alliances is held constant, just as we controlled for the influence of alliances in assessing the effects of democracy.

There is substantial evidence that the peace-inducing effects of trade remain significant even when the reverse impact, of both conflict and the state of political relations on trade, is estimated simultaneously (Polachek 1992; Mansfield 1994; Reuveny and Kang 1996; Kim 1998). In order to get the temporal sequence right in the analyses we report, we explain conflict in one year by referring to conditions, namely, the level of trade, in the previous year. This precaution is particularly important in evaluating the effects of trade on interstate conflict and helps us avoid clearly erroneous causal inferences. In Chapter 6, we explore further the reciprocal effect of conflict on trade with an analysis that predicts the level of bilateral trade from looking at conflict in the previous year and other influences.

We adopt one other method in this chapter in an effort to get the causal inference correct. We look at the effect of a country's total trade, that is, its exports to and imports from all other countries, on the likelihood of military conflict. In the discussion so far, we have emphasized the role that the trade between two states is expected to play in their relations with one another. That is where the effects of trade on conflict should be most evident. But the classical liberals thought that states that were inte-

grated into the world economy would be peaceful. Economic openness, as indicated by the ratio of a state's total trade to its GDP, should also constrain the use of force, even against another state with which bilateral trade is limited. This is because an ongoing dispute is apt to discourage some traders and investors from third countries from engaging in economic activity with the disputing states. States with open economies must be concerned about these indirect costs of resorting to military measures. Consequently, the total trade-to-GDP ratio, as well as states' dependence on bilateral exchange, indicates the costs associated with the use of force. This provides a valuable test of the causal influence of economic interdependence on the likelihood of conflict because it is difficult for a state to manipulate the economic importance of its trade with all countries simultaneously. It is relatively easy for a state to restrict its bilateral trade with a potential adversary while increasing its commercial relations with others, but it would be much more costly to reduce trade with all states in anticipation that conflict were imminent. National economies are not so easily restructured.

For example, by the end of the nineteenth century, Great Britain worried about the effects that conflict might have on its trade and on its economy. Having adopted the policy of free trade, with few artificial supports for domestic agriculture, Britain had become largely dependent on imports of food to feed its people and on the export of manufacturers to pay for those imports. This dependence on trade—and its vulnerability to disruption—meant that a protracted war in Europe would seriously hurt the standard of living of the British people. Britain's leaders must have been mindful of this danger. Russia, by contrast, was much more self-sufficient in this period because of its great continental territory and had less to fear from a disruption of foreign trade.

The vulnerability of a state that is dependent on trade was illustrated more recently, when in 1996, Taiwan too openly asserted its independence from the Chinese government on the mainland. In retaliation, Beijing fired missiles into the ocean near Taiwan. This did not so much disrupt commerce between the two countries, which was not very great, as frighten Taiwan's important trading partners. The possibility of conflict in the Taiwan Strait made that commerce much more risky and costly, for both Taiwan and its commercial partners, as shipping and insurance costs rose sharply. Because Taiwan is heavily dependent on trade with many countries, this proved to be a significant threat.

Much of the theorizing in the nineteenth century about the influence of trade on interstate relations centered on the effects of openness to the global economy generally rather than on the effects on bilateral relations. The liberals advocated free trade because it would lead to specialization according to comparative advantage and higher levels of international trade and investment. This would encourage economic growth and create in each state a large political constituency for maintaining the interdependent global system. All this necessitated peace. Thus, high levels of economically important trade were expected to create broad commercial interests that would encourage peace with everyone, not just with a state's closest trading partners. As David Ricardo put it, "The effects of war may so raise the freight and insurance on its conveyance, that it can no longer enter into competition with the home manufacture of the country to which it was before exported. In all such cases, considerable distress, and no doubt some loss, will be experienced by those who are engaged in the manufacture of such commodities" (Sraffa 1951, 255). Thus, in the analyses we report below, we consider the effect of economic openness, as indicated by the ratio of state's total trade (exports and imports) to its GDP, as well as of bilateral interdependence, on the likelihood of interstate conflict.

Testing the Effects of Trade

To sort out the evidence for these varied claims about the effect of trade requires analyses like those in the previous chapter. We will look at the same set of politically relevant dyads over the years 1885–1992, using the same method of logistic regression analysis. This means that we explain the probability of a dyadic dispute with the same control variables—a measure of the distance separating the members of a dyad, an indicator of whether they are contiguous states or not, and one showing whether the dyad contains only minor powers—and the same policy-relevant influences—level of democracy ($DEMOC_L$), whether the pair was allied, and a measure of the bilateral balance of power.⁴ To these we add trade

⁴We omit the higher democracy score ($DEMOC_H$) from the analyses in this chapter because it did not prove to be statistically significant. Adding it makes little difference to the effects of the other variables considered below.

and economic growth. As before, we lag all explanatory variables by one year to give us some protection against confusing the direction of any causal relationship; so the first year of disputes we explain is 1886.

Our hypothesis, derived from Kant and other classical liberals, is that the probability two states will become embroiled in conflict is inversely related to the degree to which they are economically interdependent. Interdependence both raises the economic interest countries have in continuing peaceful exchange and provides a medium of communication that can be useful in preventing or resolving disagreements short of violence. For the post–World War II era, we use International Monetary Fund data regarding the direction and value of trade as the basis for our measure of bilateral economic interdependence (IMF 1997). The IMF reports reasonably complete information on its member countries' exports to and imports from their trading partners.⁵

Information about international trade becomes more problematic

⁵The IMF does not report the value of trade for all possible pairs of states; data are sometimes missing. IMF members sometimes report that the value of their trade with a particular country is zero, but often they do not report the absence of trade. This is true both for small, distant countries and for dyads involved in long-term rivalries, such as many of the Arab-Israeli dyads. Syria, for example, does not report that it had no trade with Israel. It makes no mention of Israel at all in its report to the IMF, unwilling even to acknowledge the existence of the Jewish state. For both types of dyads, no entry in the IMF statistics really indicates that no significant trade took place. To minimize the loss of cases due to missing trade data, we first supplemented the IMF statistics with data collected by Katherine Barbieri, which are available at http://pss.la.psu.edu/TRD_DATA.htm. Then we assumed that trade data for IMF members that were still missing indicated zero (or near zero) trade. According to an official in the IMF's Statistics Department, this is a reasonable assumption because IMF members are required to report trade and a variety of other statistics to the IMF by the terms of their agreement with the fund. If neither member of a dyad belonged to the IMF, we made no assumption about their level of trade. Most notably, this meant the loss of about 2,500 observations (dyad-years) involving two Communist countries during the cold war. Many dyads involved in the Korean and Vietnam Wars are omitted because of missing data on the gross domestic products of the Communist states. As shown in Oneal and Russett 1999a, evidence for the pacific benefits of trade does not depend upon assuming that missing data means that there was no trade. In our earliest analysis (Oneal et al. 1996), we used other assumptions to reduce the problem of missing data and there, too, found strong support for the pacific benefit of trade.

when we move to the years before World War II. The League of Nations contemporaneously compiled data on bilateral trade in current values, along with exchange rates, during the years 1920–38 (League of Nations, various issues). While the accuracy and comparability of these data are undoubtedly not as good as the later IMF reports, they are the best available. There are no institutional compilations of trade data for the years of the two world wars or for the period before 1914. Before World War I, the annual editions of *The Stateman's Yearbook* (Epstein 1913 and other editors and years) are the closest approximations, but the data there are less standardized, the appropriate exchange rates for converting them to a common currency are less certain, and there are more missing data. So we took several steps to minimize the effect of missing data and to insure that what we had was as accurate as possible.⁶

We expect trade to influence dyadic relations to the degree that it is economically important. Only then will the economic agents involved—the exporters and importers, shippers, and consumers—be politically powerful and motivated to influence national leaders. To calculate the economic importance of trade, we divided a country's trade (exports plus imports) with its dyadic partner by its gross domestic product. This gives

⁶We took several steps to minimize the effect of missing trade data in the 1885–1949 period. We used information about one state's exports to another to infer its partner's imports. We collected estimates from other sources, compared them to the data from *The Stateman's Yearbook* and the League of Nations, and adjusted the data from our principal sources as appropriate. These sources include Mitchell 1981 for Europe and other volumes by Mitchell for all other regions of the world and for the United Kingdom; U.S. Department of Commerce, *Historical Statistics of the United States: Colonial Times to 1970*; and Katherine Barbieri's data. Exchange rates come from U.S. Federal Reserve Bank sources, *The Stateman's Yearbook*, and Global Financial Data Company (www.globalfindata.com). We also interpolated between known values of trade and used the average value of a dyad's trade to extrapolate. Finally, if neither state in a dyad reported exports to or imports from the other, we assumed that there was no trade between them. We conducted several tests to see if these methods might have biased our results. First, we dropped all zero values of trade, and then we dropped all interpolations and extrapolations. Analyses with the remaining "real" data for the pre-World War II period were consistent with those discussed in the text.

us a measure of the degree to which it is economically dependent upon this bilateral commerce (DEPEND).⁷

We must acknowledge that dyadic trade, even when adjusted for the size of the overall economy, is an imperfect indicator of economic interdependence. For one thing, the composition of trade is not considered. A country like Japan that imports large quantities of oil and food, for example, may experience greater vulnerability than our measure of dependency indicates. Yet to the extent that international prices reflect the true value of commodities, including the possibility of disruptions to existing channels of supply, the dyadic trade-to-GDP ratio will accurately measure a country's dependence on its trading partner.

It would also be good if we were able to analyze the effect of foreign investment on the likelihood of conflict. This type of interdependence has become increasingly important in recent decades. Between 1970 and 1997, worldwide GDP more than doubled in constant dollars, and trade quadrupled, but foreign direct investment expanded to more than 700 percent of what it had been (Rosecrance 1999, 37). Foreign investment and the globalization of production, like trade, should increase the incentive for peace. Investment creates similar networks of shared interest and communication. Military conflict raises the risk that foreign investments will be expropriated or destroyed. Unfortunately, dyadic investment data comparable in coverage to the trade data simply do not exist; they are particularly sparse and unreliable prior to the 1950s. Yet trade and foreign investment are highly correlated, so consideration of this important influence is not completely absent from the analyses we conduct. It is true that trade and foreign investment are to some degree substitutes: a manufacturer may export goods to another country rather than make them there. But even traditional forms of trade often involve the establishment of for-

⁷We used GDPs in current dollars to be consistent with the trade data. We began with the estimates of constant dollar GDP in Summers et al. 1995 for the years after 1949 and with Maddison's estimates (1995) for fifty-six countries in all regions of the world for 1870–1992. We regressed those estimates on year and COW's estimates of annual energy consumption to predict data for additional years and additional countries. Energy consumption is a good correlate of incomes, as Morgenstern, Knorr, and Heiss 1973 note (see also Oneal 1989), but the efficiency with which energy is converted to useful output (GDP) varies through time. We converted these constant dollar GDPs to current dollars using Maddison 1991b.

eign commercial operations. Increasingly, however, trade takes place within multinational corporations. Some 40 percent of all merchandise trade involves transactions between subsidiaries of the same company (Alworth 1988, 208; Spero 1990). We can be reasonably confident, therefore, that our trade measure reflects this important dimension of international economic relations, too.

We assume that the state with the lower bilateral trade-to-GDP measure is the one less constrained from using force and, therefore, that it has the greater influence on the likelihood of dyadic conflict. This state has greater freedom to initiate violent conflict because its economic costs and the beneficial effect of communication would be less. An example is provided by events in Guatemala in 1952. The leftist government of Jacobo Arbenz seized vast areas of agricultural land, including a quarter of a million acres belonging to United Fruit, an American multinational corporation. Only small sums were paid for the land, which was then turned over to landless Guatemalan farmers. The economic stakes for the United States were small relative to its huge economy, so there was no significant constraint preventing the Eisenhower administration from adopting a tough line. Severe economic sanctions were imposed on Guatemala. The Guatemalan government fell increasingly under Communist influence, and in August 1953, the U.S. National Security Council defined the situation, which took place at the height of the cold war, as a threat to national security.

Arbenz tried to avoid worsening his relations with the United States, but diplomatic conditions deteriorated, and the U.S. government increased its military assistance to a group of rebels, led by Castillo Armas, which operated from bases set up with CIA support in Nicaragua. In January of 1954, Arbenz made arrangements to buy arms from Communist Czechoslovakia. In turn, the United States set up a naval blockade of Guatemala to stop arms shipments from Eastern Europe, concluded a military assistance agreement with Guatemala's neighbor Honduras, and began shipping tanks and planes to Honduras to support an invasion. In June, with heavy covert support from the CIA and the Pentagon, Armas invaded Guatemala and the Arbenz regime collapsed (Cullather 1999). Although the Arbenz government certainly provoked U.S. displeasure, it was the United States that militarized the dispute: supporting Armas and his rebels, initiating the blockade, and ultimately encouraging the invasion. While Washington was not prepared to commit the U.S. Army to

the effort, the invasion could not have occurred without U.S. support. The United States was far less constrained from using force than Guatemala was.

This example illustrates why we expect the less constrained state in each dyad to be primarily responsible for the presence or absence of conflict between them. It is the weak link in the chain of peace. Accordingly, we include in our statistical analyses the bilateral trade-to-GDP ratio for the state with the lower dependence score ($DEPEND_L$). To determine if asymmetric economic relations increase the probability of conflict, as dependency theorists suggest, we can add a second trade variable ($DEPEND_H$) to our model of conflict. If one country is much more dependent on existing commercial relations than the other, exploitation (or a perception of exploitation) may make political relations subject to conflict, or it may at least reduce the benefits of economic exchange from what they would be if their trade were equally important to both and the two states were truly interdependent. Since the numerator (dyadic trade) is the same for both countries, the state with the smaller economy will have the higher trade-to-GDP ratio. The difference between $DEPEND_L$ and $DEPEND_H$, as in the case of the United States and Guatemala in 1953, can be very great.

The recent trend in economic relations, as well as their most recent level, may also affect the likelihood of military conflict. To measure the trend in relations, we calculated the change in the bilateral dependence of states over a three-year period, from four years prior to the current year, when the state of dyadic relations (peace or a dispute) is being assessed, to the year just before a possible conflict. A decline in the economic importance of bilateral trade should be associated with an increase in the likelihood of dyadic disputes. On the other hand, a rising level of trade may signal improving interstate relations. The relatively long four-year span helps to maximize the chance that a decline in trade may be more a cause of worsening political relations than simply a near-term reflection of an anticipated violent conflict.⁸ We assess the consequence of change in the economic importance of trade by including in our analyses the magnitude of the change in the trade-to-GDP ratio (negative or positive) for whichever state in each dyad experienced the greater change.

⁸To minimize the loss of cases, we substituted the change in trade over a three- or two-year span when values were missing at the beginning of a dyadic time series.

We also consider the effect of total trade on the likelihood of a dispute in order to take into account the economic effects of conflict on third parties. Economic openness, a state's trade with all countries as a fraction of its GDP, is a measure of a state's interdependence in the global economy generally. As with Britain in the late nineteenth century or Taiwan in the twentieth, open economies are subject to disruption by wars and rumors of war (Rosecrance 1999). We measure openness (OPEN) as a country's total exports plus its total imports divided by its GDP.⁹ We expect the less open state (OPEN_L) to be less constrained from resorting to violence because its economy is less subject to the disruption that hostilities might cause.

The degree to which a country depends upon international trade is influenced by the size of its domestic market. Small countries tend to be more open than larger ones. A small country will find it difficult to be highly self-sufficient. It will need many goods and services that it cannot produce efficiently on a small scale, so it will import them. To be able to afford these imports, it will have to specialize in making goods that it can produce relatively efficiently and sell abroad what it does not need for its own consumption. Consequently, small countries have little choice but to concentrate on the export of natural resources (oil, mineral, specialized agricultural products) if they have them (as does Kuwait) or on a set of high-tech manufacturers or services for which they possess a comparative advantage (as does Singapore). Big economies, with internally varied climates and resource endowments and a large number of consumers, can produce a wide range of products at close to the prices that prevail in the world market, so they are usually more self-sufficient.

Some states make a greater political commitment to the liberal principles of free trade and comparative advantage than do others, so the correlation between economic size and openness is far from perfect. At the end of the nineteenth century, Great Britain had a substantially more open economy than its competitor Germany, although they were comparable in size. In 1968, the United States, with a GDP more than twice that of the Soviet Union, imported and exported goods equal in value to 8 percent of its GDP, whereas the openness of the Soviets was under 3 percent.

⁹Our data for total trade for 1950–92 are from Summers et al. 1995. For the years before 1950, we relied primarily on the volumes by Mitchell identified above.

For West Germany and Japan in the same year, then with very similar GDPs, the trade percentages were 33 percent and 18 percent, respectively. Thus, the relative impact of the international economy on a state's politics depends not just on the size of its GDP but also on other economic and political factors, including the policies adopted by the national government. This is fortunate because we want to be sure we are measuring economic interdependence, a liberal variable, and not size, which is associated with national power and the argument of the realists.

Trade Does Reduce Conflict

We are now ready to evaluate the liberal thesis that trade reduces the likelihood of military conflict. First, we consider the consequences of economically important bilateral trade by adding the bilateral trade-to-GDP ratio for the larger and thus less dependent state to the basic regression equation we used in the last chapter. Thus, the variable $DEPEND_L$ is combined with our measure of democracy and the various realist variables. We use the entire 1885–1992 set of politically relevant dyads. Then, in turn, we ascertain the effects of asymmetric trade (as indicated by $DEPEND_H$), of openness, and of growth in dyadic trade. As in the previous chapter, we evaluate the effects of each of the theoretically interesting variables by first setting them all to their baseline values and then estimating the likelihood of a dispute for this "typical" dyad. Thus, we make the measure of democracy equal to 0, assume the dyad is unallied, and set each of the continuous variables at their means for the contiguous subset of dyads.¹⁰ Then, one by one, we add one standard deviation to each of the risk factors while holding the others constant. This tells us the independent risk-reducing effect of that variable. This information is reported in column 1 of Table 4.1.

These results provide strong support for the liberal peace. Both democracy and economically important trade are strong and statistically significant constraints on the use of force. The likelihood of a dispute is much lower when states are dependent on bilateral trade or are democratic.

¹⁰We use the median of $DEPEND_L$ as its baseline value. It is more representative than the mean because of the skewed distribution of this variable.

Table 4.1: Percentage Change in Risk for Annual Involvement in a Militarized Dispute, 1886–1992: Contiguous Dyads, Realist Variables, Democracy, and Economic Interdependence

	(1) <i>With Dyadic Variables</i>	(2) <i>Adding Openness</i>
<i>All variables at baseline values except:</i>		
ALLIES equals 1	–47%	–35%
POWER RATIO increased by one standard deviation	–34	–33
DEMOC _L increased by one standard deviation	–36	–31
DEMOC _L decreased by one standard deviation	+54	+44
DEPEND _L increased by one standard deviation	–44	–35
OPEN _L increased by one standard deviation		–27
DEPEND _L and OPEN _L both increased by one standard deviation		–52

Both influences are substantial despite the fact that democracies tend also to have higher levels of trade than other states. Controlling for the benefits of interdependence, democracy still reduces the probability of a dispute by 36 percent. The impact of the bilateral trade-to-GDP ratio is even bigger: an increase of one standard deviation makes a dispute 44 percent less likely. There is less than one chance in a thousand that the close association between either interdependence and peace or democracy and peace would have occurred by chance. Kant and the other classical liberals were right: both democracy and trade increase the prospects for peace. The other variables perform as expected: alliances are strongly associated with less dyadic conflict, and an overwhelming imbalance of power (POWER RATIO) still has an important deterrent effect. The reductions in risk associated with these influences are 47 percent and 34 percent, respectively. All the variables (including distance, contiguity, and minor power dyads, which are not shown in the table) are statistically significant at a very high level (.001).

The pacific benefits of economically important bilateral trade seem

well illustrated by the experience of the United States with China over the past twenty years. After the Communist government began to open its economy in the late 1970s, its political relations with the United States became far more peaceful than they had been during the cold war. This thaw in relations began with a deliberate political decision to improve them, but as trade increased, both sides gained a greater stake in keeping them peaceful. This happened despite the fact that China did not become significantly more democratic.

Next we assess the view of dependency theorists that asymmetric dependence on trade reduces the pacific benefits of commerce. Asymmetry is said to give a powerful state the opportunity to exploit its trading partner. This may provoke disputes as either the disadvantaged state rebels or the powerful state uses military force to enforce its advantage. The example of U.S.-Guatemalan relations in the early 1950s might be an example of a general tendency. To test whether trade between a large and a small state has different implications for interstate relations than trade between states of equal size, we add to our equation the higher bilateral trade-to-GDP ratio (DEPEND_H) in each pair of states. The coefficient of this variable will be positive if, holding the trade-to-GDP ratio of the bigger state constant, greater dependence on the part of the smaller state increases the likelihood of interstate conflict. This analysis is similar to the one in the preceding chapter where we asked whether autocracies and democracies are particularly prone to fight.

Contrary to the expectation of dependency theorists, the benefits of trade do not depend upon the states' being of similar size. The estimated coefficient of DEPEND_H was not significantly different from zero. This means that the benefits of trade are not importantly affected by the higher trade-to-GDP ratio, only the lower one. Indeed, the coefficient was negative, not positive. What little effect an unequal balance of trade has is to reduce the likelihood of conflict further. Economically important trade between large states and small states increases the prospects for peace just as it does for states of equal size.

We can only speculate why this is true. One possibility is that the economic domination of small states by big trading partners is so complete that acts of resistance are not undertaken. If the political system of a small trade-dependent state is heavily dominated by an elite (such as plantation owners or mining companies in Central American countries) trading with

the big country, that state's restraint in avoiding overt conflict may deter it from any actions that would cut off profitable trade. If the small country does not resist the powerful one, there may be no need for the latter to exercise its military might.

The alternative explanation, and the one that would be emphasized by liberals, is that trade really does significantly benefit both partners. Trade is not forced. It is voluntary. If a buyer and a seller agree to a transaction in a free market, it is because both expect to be better off than they would have been without the exchange. If both are better off, neither one would prefer to see the relationship disrupted by hostilities. We cannot settle this debate here, but this second explanation coincides with doubts that the international economy works to the disadvantage of developing countries (de Soysa and Oneal 1999).

Are Open Economies More Pacific?

Next we ask whether pacific benefits also accrue from a high level of economic openness generally, or whether it is only bilateral trade that affects the likelihood that two states will become involved in a militarized dispute. We report in column 2 of Table 4.1 the results of estimating the independent effect of high levels of total trade, controlling for bilateral interdependence, democracy, and the realist variables. Adding openness ($OPEN_L$) to our model of interstate conflict provides strong support for the general thesis that economic interdependence reduces the likelihood of military conflict. All of the variables that had a statistically significant impact in the analysis reported in column 1 still do. As before, all the variables are significant at the .001 level, meaning that there is less than one chance in a thousand that such strong correlations would be found if the information used was just a randomly generated series of numbers.

Even when both dyadic trade and general openness are included in the same analysis, each makes a substantively important independent contribution to reducing the probability of a militarized dispute. Again, dyadic trade has the greatest impact, reducing the risk of a dispute by 35 percent, as much as if the two states were formally allied. A higher level of democracy reduces the likelihood of conflict by 31 percent, but openness to the world economy also has a notable impact: an increase of one standard de-

viation reduces the risk of a dispute by 27 percent.¹¹ The impact on the likelihood of conflict of democracy, bilateral trade, and the existence of an alliance all fall slightly because these variables are correlated with one another. That is, states that are democratic trade more with one another, tend to be allied, and have more open economies generally.¹² The most important thing to note is that, despite these interrelationships, the independent effect of each variable is still quite dramatic.

It is not surprising that bilateral economic relations are a better predictor of the tenor of dyadic relations than openness is, but it is encouraging that total trade also has an independent, statistically significant, and substantively important impact. Countries that are open to external economic relations are constrained from using force even against rivals with whom commercial ties are limited. It is worth estimating the combined effect of high levels of dyadic trade and openness. Because we are estimating the probability of a dispute, which must lie between 0 and 1, their combined effect is not as big as adding 35 percent and 27 percent together. Nevertheless, if both dyadic trade and openness are increased above their baseline levels, the likelihood that two countries will experience a dispute falls by more than half (52 percent). It is the isolated countries of the world that represent the greatest danger to peaceful international relations.

We also looked at whether the trend in bilateral trade dependence affects the probability of conflict. To do this, we performed new analyses that included measures of the level of economic interdependence and of the change in the level over the previous four years. There is no need to present these results in a table; they are easily summarized. Contrary to

¹¹As noted earlier, smaller states tend to have more open economies than large states. To make sure that our measure of openness was not acting as an additional measure of national power, rather than interdependence, we conducted an analysis adding the ratio of the larger state's GDP to the smaller state's GDP to the model. If openness were really indicating just economic size, this measure would have been negatively related to the onset of conflict, as is the *POWER RATIO* derived from the COW data. In fact, the GDP ratio had the wrong sign and was far from statistical significance; $OPEN_L$ remained significant at the .002 level.

¹²A country's total trade is, of course, the sum of all its bilateral commercial exchanges, but $OPEN_L$ and $DEPEND_L$ are only moderately correlated ($r = 0.33$) since trade can be shifted from one country to another while maintaining the same level of total exports and imports.

our expectations, we found no significant evidence that the trend in trade makes a difference in the likelihood of dyadic conflict. We measured change in economic relations in several ways. None of them indicated that falling levels of trade are an additional portent of danger or that rising levels per se reduce the probability of a dispute. In these analyses, it was the level of trade, not recent changes in it, that proved useful in accounting for interstate violence.

This does not mean that falling levels of trade are unimportant. If economic interdependence declines, the prospects of conflict do increase, because the level of trade is closely associated, as we have seen, with the incidence of conflicts. It is just that low and declining trade does not herald greater danger than does trade that has been low over all of the four most recent years. The Great Depression, as we briefly noted earlier, is a dramatic reminder of the dangers that attend economic crises. In the 1930s, there was a sharp decline in living standards and then an even sharper drop in the volume of international trade, as states imposed new tariffs and other trade restrictions to protect their domestic markets from foreign competition. These "beggar-my-neighbor" policies invited retaliation in kind. As the retaliatory policies took effect, states became less dependent on maintaining good political relations with each other. This led to a downward spiral or vicious circle of hard economic times, failed democracies in much of Europe and in Japan, and darkening clouds of war. International organizations, notably the League of Nations, were too weak to act effectively. Interdependence reduces the prospects of violent international conflict. If the links of interdependence are broken, however, so too are important constraints on conflict.

As with the democratic peace, one of the key questions about the consequences of trade is whether the benefits of interdependence are confined to the post-World War II period. It is often noted, for example, that World War I occurred at a time when the countries of Europe were more interdependent than they had ever been before and more interdependent than they would be until many years after the end of World War II. We will look at this matter in detail when we consider the effects of the full set of Kantian influences, including international organizations, in the next chapter. For now, we will say only that the benefits of trade have been greatest in the years after 1950, but they were substantial for the years 1886–1939 as well. We do not find any reason to think that the liberal peace is limited to the cold war. As we noted in the last chap-

ter, the biggest difference between the cold war and earlier periods was that alliances became associated with reduced conflict only after World War II.

Economic Growth and Conflict

Our consideration of the influence of economic conditions on the likelihood of international conflict should not stop with trade. The role of economic growth also warrants attention. States with strong economies, those that enjoy prosperity and are experiencing economic growth, may be disinclined to fight. Their populations are likely to be satisfied with the economic and political status quo, and as liberals have emphasized, violent conflict is inconsistent with many financial and commercial relations. One reason why the rich industrialized countries gave up wars of imperialism in the latter part of the twentieth century was that their prosperity and continuing growth made such wars look unattractive from any reasonable cost-benefit perspective (Mueller 1989). The higher the rate of economic growth, presumably the greater the popular satisfaction and the less inclined the people should be to engage in militarized conflicts.

Moreover, leaders do not have an incentive to start a conflict if the economy is prospering. In Chapter 2, we discussed the substantial literature on the hypothesis that some wars stem from an effort by leaders to divert attention from domestic difficulties, such as a failing economy, toward foreign adversaries (or scapegoats). Presumably, the greater is the rate of economic decline, as indicated by falling personal incomes, the greater will be the incentives for leaders to divert attention from economic conditions. The evidence for that effect from previous research is mixed, as we noted, but the hypothesis is interesting and plausible, so we want to explore the possibility here. Both these considerations—about the conflict-mitigating effect of economic growth and about the conflict-inducing effect of economic decline—suggest that the state with the lower rate of economic growth will be the greater danger to peace.

Although the diversionary theory of war seems a plausible account of some individual cases, such as the Argentine invasion of the Falkland Islands in 1982, the effect that we should expect the economy to have on the likelihood of conflict is not entirely obvious. Some have suggested that economic growth *increases* the prospects of war and that economic

decline reduces it. If a country's economy is deteriorating, its industry may be incapable of producing the war material necessary to support an extended conflict with another country. Or domestic dissatisfaction with the government could be so serious that the public will not support it in a foreign dispute. It is commonly believed that the war with Germany in 1914–17 only added to the dissatisfaction of the Russian people with the tsar; it did not divert attention away from the economic suffering and instead provided an additional reason for a revolutionary explosion.

Rapid economic expansion, on the other hand, may provide the “fuel,” in the form of greater material capabilities, for military adventures and may encourage a general sense of optimism about the strength of the country that would embolden them. To this could be added the hypothesis that rapid economic growth intensifies the need for imports of vital commodities. Both Hobson's and Lenin's theories of imperialism, formulated at the height of the era of imperialism at the turn of the last century, stressed the danger of international conflict inherent in efforts to acquire colonies in a diplomatic and military competition with other imperial powers. This idea was taken up by later social scientists (Choucri and North 1975; Goldstein 1988; Pollins and Schweller 1999), who argue that economic growth creates “lateral pressure” that forces states to look abroad for resources, raising the danger of war as they encounter others on the same quest. These various considerations agree in predicting a greater danger of conflict with greater economic growth and less conflict when states are in decline.

When faced with such conflicting predictions about the effect of economic growth on conflict, empirical analysis is particularly important. To evaluate the rival hypotheses, we used information regarding personal incomes—real GDP per capita—to calculate economic growth rates over three years. The term “real” indicates that the effects of inflation have been factored out and that the variability of prices in different countries is also taken into account; thus, we are assessing change in the average person's actual standard of living. The initial hypothesis was that a strong economy discourages conflict and economic decline encourages it. We conducted a couple of tests of this view. In the first, we assumed that the likelihood of conflict depended mainly on actions by the less constrained state in each dyad. According to liberals and those who think wars occasionally serve a diversionary purpose, this would be the state with the lower rate of growth. Thus, in our first test, we used the variable

GROWTH_L. As in our other tests, we sought to insure that the disputes we were trying to explain had not influenced the variables we were using for that very purpose, so we used the lower rate of growth in each dyad over the three years *preceding* the year in which conflict might have occurred.

We wanted to consider carefully the experience of both members in each dyad, so we also conducted a second test. In this analysis, we created one variable that recorded the faster rate of growth in each dyad and a second that measured the severity of the worse decline.¹³ In this way, we hoped to capture the effect of growth on interstate conflict in more complex situations, for example, if one member of a dyad is growing rapidly while the other is experiencing a declining standard of living. These variables also allowed us to investigate the alternative hypothesis that economic growth creates wealth that enables states to act more aggressively and that a declining economy limits the use of force.

However we measured the effect of growth, we found no significant influence on the occurrence of militarized disputes. There seems to be no systematic relationship between conflict and either a successful, growing economy or economic decline. Our measures of growth in both tests never approached statistical significance. Consequently, we do not report any effect of this factor on the risk of disputes. This evidence suggests that *economic growth is neither the enemy of peace nor essential to it*. More important are the ties of interest and communication that derive from substantial commercial transactions with other states, the character of states' political regimes, whether they are allied, the balance of power, and the other influences we have been discussing.¹⁴

¹³Heldt (1999) suggests that the effect of growth be assessed in this way. One variable indicates the higher rate of growth in the dyad if one or both states experienced growth at a positive rate. If both states experienced a decline in their per capita GDPs, this variable equals 0. Similarly, the other variable equals the change in GDP per capita of the state that experienced the greatest economic decline. If both states experienced actual growth in incomes, this variable is set equal to 0. Heldt also asks whether democracies are more prone to use force under various economic conditions than autocracies are, but he finds no systematic effect.

¹⁴Mousseau (2000) and Hegre (2000), however, suggest that democracy and economic interdependence may have little dispute-reducing effect for very poor countries, such as Bangladesh.

Economic Interdependence and Peace

We can now summarize the results we have reported in this chapter. They are important and encouraging. There is strong, consistent evidence that economic interdependence, like democratic institutions and norms, significantly reduces the risk that two states will become involved in a militarized dispute. Over the years, liberals have claimed that democracy and free trade not only increase individual liberty and prosperity but also ameliorate international conflict. Our analyses indicate they have been right: the pacific benefits of both democracy and economic interdependence are evident and substantial. In addition, neither variable eliminates the importance of the other. Higher levels of economically important trade, as indicated by the bilateral trade-to-GDP ratio, are associated with fewer incidences of militarized international disputes. A one standard deviation increase in the bilateral trade-to-GDP ratio reduces the annual probability of a dispute more than one-third below the baseline rate. Such a powerful benefit appears even when we control for a variety of potentially confounding, theoretically interesting influences, such as geographic contiguity, the balance of power, alliance bonds, democracy, and economic growth rates. Nor are the pacific benefits of trade reduced by asymmetric economic relations, as dependency theorists have feared.

Economic openness (the total trade-to-GDP ratio) is also associated with a reduced risk of conflict. Even when we hold constant the effects of bilateral interdependence, a one standard deviation increase in openness makes a dispute 27 percent less likely. If both bilateral trade and openness are increased, the likelihood of dyadic conflict drops by 52 percent. This evidence for the pacific benefits of openness is important for two reasons. First, it indicates that states recognize the consequences of militarized disputes for their economic relations with third parties. As a result, even when bilateral trade and investment is limited, states can be constrained by wider economic forces from taking military action. Second, the significance and substantive importance of openness as a predictor of peaceful relations increases our confidence that economically important trade affects the probability of conflict and is not just a consequence of peaceful relations. Deteriorating political relations might cause a state to reduce its economic dependence on a potential adversary, but it is much harder for a state to manipulate its total trade-to-GDP ratio by restricting economic ties with all states simultaneously. Our analyses with openness do not dis-

pel all doubts about the causal influence of trade on conflict. We want to get a better indication of the strength of the influence of conflict on trade, and so we return to this issue in Chapter 6 when we investigate the influences that shape bilateral trade.

We found no support for some other hypotheses. The short-term trend in bilateral interdependence has little impact on the likelihood of conflict. We found no evidence that change in the level of the bilateral trade-to-GDP ratio indicates states' expectations regarding the future of their relations or that declining trade exacerbates interstate relations. It is the economic importance of current levels of trade that affect the prospects for peace, not change in the level. Nor did we find a systematic relation between economic growth (or decline) and states' involvement in militarized disputes with each other. Neither liberals, who think growth might have significant benefits, nor those concerned that growth might trigger conflict as states compete for resources can find support here. Nor is there evidence that the leaders of states are inclined to divert attention from economic difficulties by engaging in foreign military adventures. Perhaps the benefits are too uncertain.

Overall, evidence we have presented in this chapter provides a strong indication that both elements of the liberals' agenda—interdependence and democracy—make independent contributions to the prospects for peace. Indeed, the pacific benefits of trade are not limited to the countries normally considered liberal—the Western democracies—or relations among them. *Countries that are interdependent bilaterally or economically open to the global economy, whether democratic or not, have an important basis for pacific relations and conflict resolution.* Still, as the liberal theorists anticipated, those that are democratic, interdependent, and economically open—as are the economically advanced democracies of the West—are most likely to be at peace.

Table A4.1: Logistic Regression of Democracy, Trade, and Other Influences on Involvement in a Militarized Dispute, 1886–1992

<i>Variable</i>	<i>(1) With Dyadic Liberal Variables</i>		<i>(2) Adding Openness</i>	
	<i>Coefficient (Standard Error)</i>	<i>Significance Level (One-Tailed Tests)</i>	<i>Coefficient (Standard Error)</i>	<i>Significance Level (One-Tailed Tests)</i>
Alliances (ALLIES)	-.677 (.152)	<.001	-.562 (.160)	<.001
Power ratio (POWER RATIO)	-.296 (.043)	<.001	-.290 (.047)	<.001
Lower democracy (DEMOC ₁)	-.0670 (.0098)	<.001	-.0564 (.0099)	<.001
Lower dependence (DEPEND ₁)	-.56.3 (14.3)	<.001	-40.6 (12.5)	<.001
Lower openness (OPEN ₁)			-1.83 (.51)	<.001
Noncontiguity (NONCONTIG)	-1.04 (.168)	<.001	-1.00 (.17)	<.001
Log distance (DISTANCE)	-.373 (.067)	<.001	-.376 (.068)	<.001
Only minor powers (MINORPWS)	-.698 (.182)	<.001	-.473 (.182)	<.005
Constant	-.580 (.533)	<.14	-.189 (.572)	<.38
Wald Chi ²	221		232	
P of Chi ²	<.001		<.001	
N	39,988		36,621	